
Project Proposal Part 1

Project Proposal & Data Identification

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Introduction

Amazon's Last mile Delivery network represents a pivotal aspect of contemporary e-commerce logistics, serving as the critical link between digital storefront and end consumers. The rapid growth of online retail has intensified consumer expectations, demanding record speed of delivery services-often as fast as same day or next day fulfilment. Amazon, a global leader in this sector, continually refines its operation capabilities, employing advanced technological fleet management to enhance customer satisfaction and maintain competitive lead.

The data given, such as extensive agent demographics, precise geospatial routing data, environmental considerations, and performance metrics, is a great foundation for an extensive PESTEL analysis. Breaking down Amazon's delivery business into political, economic, social, technological, environmental, and legal factors will reveal key external drivers that shape operational outcomes. This analytical framework will strictly, establish potential threats, emerging opportunities, and strategic areas of influence that are critical in guaranteeing resilience and operational excellence in the extremely volatile e-commerce logistics sector.

Operational Overview:

Amazon's last-mile delivery is completed in four integrated stages:

- **Fulfillment & Dispatch:** Delivery stations within the region receive sorted parcels from nearby regional hubs.
- **Dynamic Routing:** A proprietary application draws upon real-time traffic and weather data to optimize each driver's route in real time.

- **Delivery Execution:** A heterogeneous fleet—company vans, motorcycles/scooters, and Amazon Flex contractors—offers flexibility across urban, suburban, and rural terrain.
- **Performance Monitoring:** On-time rate, delivery lead time, and customer ratings are tracked in real time to continually optimize operations.

This end-to-end process must balance speed, cost, and reliability while reacting to shifting demand peaks, last-minute order cancellations, and infrastructure constraints.

Data Cleaning Process

To prepare for rigorous analysis, we'll undertake the following steps:

- **Missing-Value Strategy:** Drop rows missing critical identifiers; impute non-critical nulls (median for numeric, mode for categories).
- **Deduplication:** Remove any repeated Order_ID entries to ensure each delivery is unique.
- **Type Enforcement:** Convert time strings to datetime objects; coerce Agent_Age and Agent_Rating to numeric, flagging anomalies.
- **Outlier Treatment:** Apply IQR-based capping to extreme Delivery_Time values.
- **Consistency Checks:** Ensure that pickup is before drop-off and GPS coordinates are within reasonable service-area bounds.
- **Standardization:** Normalize categorical columns (Vehicle, Weather, Traffic) to uniform spelling and case.
- **Feature Engineering:** Derive $\text{Duration} = \text{Delivery_Time} - \text{Pickup_Time}$ and capture hour, weekday, and season indicator.
- **Validation Log:** Document each cleaning step, including before and after record counts, to enable reproducibility.

Initial Insight: External Factors Impacting Amazon Delivery

Early Insights: External Drivers of Amazon Delivery

Political:

- Gig-economy legislation and minimum-wage regulation directly affect driver cost and availability.
- Global trade policy and tariff regimes can prohibit cross-border shipments.

Economic:

- Fuel price volatility drives rapid swings in operating cost.
- Regional GDP growth and consumer spending patterns modulate order volumes.

Social:

- Demand for ultra-fast delivery (same-day, one-day) increases logistical complexity.
- Urbanization concentrates deliveries into high-density, high-traffic zones.

Technological:

- AI-optimized routing and warehouse robotics optimize efficiency.
- Emerging drone and autonomous vehicle pilots will transform last-mile logistics.

Environmental:

- Carbon-emission targets propel the shift to electric delivery fleets.
- Extreme weather events (heatwaves, snowstorms) pose persistent disruption risks.

Legal:

- Data-privacy regulations (GDPR, PIPEDA) govern handling of customer and driver information.
- Vehicle-safety and liability regulations set fleet compliance standards.

Conclusion

This proposal lays a robust groundwork for conducting an extensive PESTEL analysis of Amazon's last-mile delivery business. Based on meticulously cleansed and enriched datasets coupled with targeted external macro-environmental information, the analysis will critically analyze the political, economic, social, technological, environmental, and legal factors influencing Amazon's operating efficiency and positioning. The conclusions of this analysis will determine principal opportunities for route efficiency improvement, fleet strategy adjustment, regulatory compliance enhancement, and environmental risk mitigation. The outcomes will lead to strategic recommendations that will enhance Amazon's operational resilience, sustain its market leadership, and enhance its responsiveness to the constantly evolving dynamics of global e-commerce logistics.